A Systematic Review of the Diagnostic Accuracy of Volatile Organic Compounds in Airways Diseases and their relation to Markers of Type-2 Inflammation

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***Equal contribution to work**

Supplementary material

1. Search strategy

A PRISMA oriented systematic search was completed from **January 1997 to December 2020** using the following sources of evidence, (i) Cochrane library, (ii) Medline and (iii) EMBASE.

Further details on methodology can be found in (PROSPERO - CRD42019141718).

Strategy 577083

#	Database	Search term	Results
1	Medline	("volatile organic compound*").ti,ab	7726
2	Medline	"VOLATILE ORGANIC COMPOUNDS"/	6691
3	Medline	(VOC OR vocs).ti,ab	7212
4	Medline	(exhal*6 OR breath*).ti,ab	113917
5	Medline	EXHALATION/	3377
6	Medline	(1 OR 2 OR 3)	14894
7	Medline	(4 OR 5)	117153
8	Medline	(6 AND 7)	1107
9	Medline	exp DYSPNEA/	19644
10	Medline	(breathless* OR dyspn*4).ti,ab	46465
11	Medline	(9 OR 10)	55084
12	Medline	(8 AND 11)	5
13	Medline	(asthma*).ti,ab	146117
14	Medline	exp ASTHMA/	121260

15	Medline	exp "PULMONARY DISEASE, CHRONIC OBSTRUCTIVE"/	50255
16	Medline	(copd).ti,ab	37549
17	Medline	(((chronic AND obstructive) AND pulmonary) AND disease).ti,ab	40946
18	Medline	exp PULMONARY DISEASE, CHRONIC OBSTRUCTIVE/	47893
19	Medline	(((chronic AND obstructive) AND airway) AND (disease OR coad)).ti,ab	5928
20	Medline	(((chronic AND obstructive) AND lung) AND disease).ti,ab	15699
21	Medline	((chronic AND airflow) AND limitation).ti,ab	1694
22	Medline	(((chronic AND obstructive) AND respiratory) AND disease).ti,ab	11050
23	Medline	(emphysema).ti,ab	21250
24	Medline	exp EMPHYSEMA/	13966
25	Medline	(chronic AND bronchitis).ti,ab	11694
26	Medline	exp BRONCHITIS, CHRONIC/	1665
27	Medline	exp PNEUMONIA/	86684
28	Medline	(pneumonia* OR lung inflammation*or respiratory tract infection* OR respiratory infection*).ti,ab	214438
29	Medline	exp "HEART FAILURE"/	110645
30	Medline	(heart failure).ti,ab	165767
31	Medline	(13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21 OR 22 OR 23 OR 24 OR 25 OR 26 OR 27 OR 28 OR 29 OR 30)	678339
32	Medline	(8 AND 31)	163

33	EMBASE	("volatile organic compound*").ti,ab	10048
34	EMBASE	"VOLATILE ORGANIC COMPOUND"/	14083
35	EMBASE	"VOLATILE ORGANIC COMPOUNDS"/	10777
36	EMBASE	(voc OR vocs).ti,ab	9092
37	EMBASE	(33 OR 34 OR 35 OR 36)	20436
38	EMBASE	(exhal*6 OR breath*).ti,ab	165810
39	EMBASE	EXHALATION/	4059
40	EMBASE	(38 OR 39)	177809
41	EMBASE	(37 AND 40)	1675
42	EMBASE	(copd).ti,ab	70312
43	EMBASE	(((chronic AND obstructive) AND pulmonary) AND disease).ti,ab	59947
44	EMBASE	"CHRONIC OBSTRUCTIVE LUNG DISEASE"/ OR "CHRONIC OBSTRUCTIVE LUNG DISORDER"/ OR "CHRONIC OBSTRUCTIVE PSEUDOEMPHYSEMA"/ OR "CHRONIC OBSTRUCTIVE PULMONARY DISEASE"/ OR "CHRONIC OBSTRUCTIVE PULMONARY DISORDER"/ OR "CHRONIC OBSTRUCTIVE RESPIRATORY DISEASE"/	112227
45	EMBASE	(((chronic AND obstructive) AND airway) AND (disease OR coad)).ti,ab	9382
46	EMBASE	(((chronic AND obstructive) AND lung) AND disease).ti,ab	24650
47	EMBASE	((chronic AND airflow) AND limitation).ti,ab	2646

48	EMBASE	(((chronic AND obstructive) AND respiratory) AND disease).ti,ab	17649
49	EMBASE	(emphysema).ti,ab	28141
50	EMBASE	exp EMPHYSEMA/	41697
51	EMBASE	(chronic AND bronchitis).ti,ab	15855
52	EMBASE	"CHRONIC BRONCHITIS"/	11611
53	EMBASE	exp ASTHMA/	237742
54	EMBASE	(asthma*).ti,ab	205990
55	EMBASE	exp PNEUMONIA/	265756
56	EMBASE	(pneumonia* OR lung inflammation*or respiratory tract infection* OR respiratory infection*).ti,ab	225025
57	EMBASE	exp "HEART FAILURE"/	443663
58	EMBASE	(heart failure).ti,ab	242128
59	EMBASE	(42 OR 43 OR 44 OR 45 OR 46 OR 47 OR 48 OR 49 OR 50 OR 51 OR 52 OR 53 OR 54 OR 55 OR 56 OR 57 OR 58)	1204355
60	EMBASE	(41 AND 59)	375

61 Cochrane search:

ID	Search
#1	"volatile organic compound*"
#2	MeSH descriptor: [Volatile Organic Compounds] this term only
#3	VOC OR vocs
#4	MeSH descriptor: [Exhalation] this term only
#5	exhal*6 OR breath*
#6	#1 OR #2 OR #3

#7	#4 OR #5
#8	#6 AND #7
#9	asthma*
#10	MeSH descriptor: [Asthma] this term only
#11	MeSH descriptor: [Pulmonary Disease, Chronic Obstructive] this term only
#12	copd
#13	(((chronic AND obstructive) AND pulmonary) AND disease)
#14	(((chronic AND obstructive) AND airway) AND (disease OR coad))
#15	(((chronic AND obstructive) AND lung) AND disease)
#16	((chronic AND airflow) AND limitation)
#17	(((chronic AND obstructive) AND respiratory) AND disease)
#18	(emphysema)
#19	MeSH descriptor: [Emphysema] this term only
#20	(chronic AND bronchitis)
#21	MeSH descriptor: [Bronchitis, Chronic] this term only
#22	{OR #9-#21}
#23	{AND #8-#22}
#24	#8 and #22

2. Risk of Bias table (QUADAS2)

Table A: Breakdown of included studies' risk of bias and applicability concerns

Asthma studies	Risk of Bias					Applicability concerns		
Author	**Patient selection**	**Index test**	**Reference standard**	**Flow and timing**		**Patient selection**	**Index test**	**Reference standard**
Olopade *et al* (1997)	Low	Unclear	Low	Low		Low	Low	Low
Paredi *et al* (2000)	Low	High	Low	Low		Low	Unclear	Low
Larstad *et al* (2007)	Low	High	Low	Low		Unclear	Low	Low
Dragonieri *et al* (2007)	Low	Unclear	Low	Low		Low	Low	Low
Fens *et al* (2009)	Low	Low	Low	Low		Low	Low	Low
Ibrahim *et al* (2011)	High	High	Low	Low		Low	Low	Low
Timms *et al* (2012)	Low	High	High	Low		Low	Low	Low
Van der Schee *et al* (2012)	High	Low	Unclear	Low		Low	Low	Low
Schivo *et al* (2013)	Low	Low	Low	Low		Low	Low	Low
Meyer *et al* (2014)	Low	High	Low	Low		Low	High	Low
Schliech *et al* (2019)	High	Low	Low	Low		Low	Low	Low
Brinkman *et al* (2019)	Unclear	Low	Low	Low		Low	Low	Low
Dragonieri *et al* (2018)	Low	Low	Low	Low		Low	Low	Low
Brinkman *et al*	Low	High	Low	Low		Low	Low	Low

(2017)								
Plaza *et al* (2015)	Unclear	High	Low	Low		Low	Low	Low
Lazar *et al* (2010)	High	High	Low	Low		Unclear	Low	Low
De Vries *et al* (2018)	Low	Low	Low	Low		Low	Low	Low
Montuschi *et al* (2010)	Low	High	Low	Low		Low	Low	Low
Brinkman et al (2020)	Unclear	Low	Low	Low		Low	Low	Low
COPD studies								
Dragonieri *et al* (2009)	Low	High	Low	Low		Low	Low	Low
Westhoff *et al* (2010)	Low	Low	High	High		Low	Low	Low
Basanta *et al* (2010)	Low	Unclear	Low	Low		Low	Low	Low
Van berkel *et al* (2010)	Low	High	Low	Low		Low	Low	Low
Fens *et al* (2011)	Unclear	Low	Low	Low		Unclear	Low	Low
Cristescu *et al* (2011)	High	Unclear	Low	Low		Unclear	Low	Low
Hattesohl *et al* (2011)	Low	High	Low	Low		Low	Low	Low
Westhoff *et al* (2011)	High	High	Low	Low		Low	Low	Low
Phillips *et al* (2012)	Low	Low	Low	Low		Low	Low	Low
Basanta *et al* (2012)	Low	Unclear	Low	Low		Unclear	Low	Low

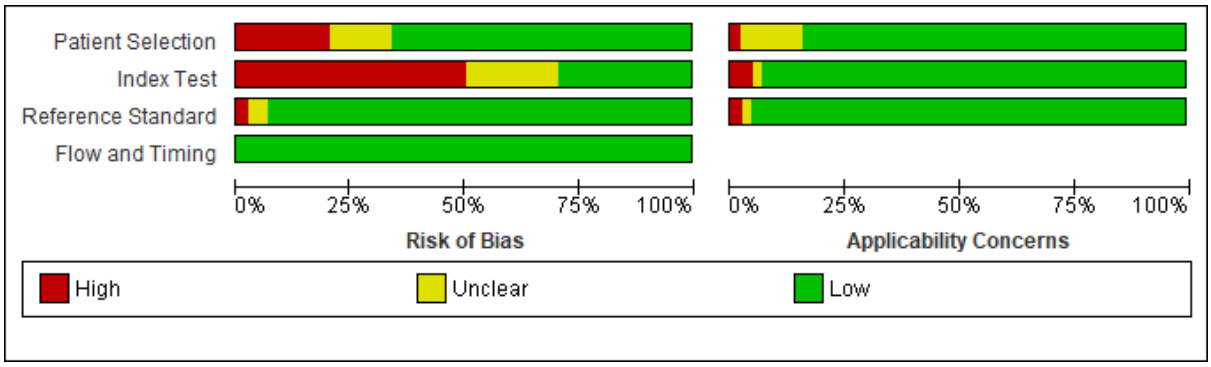
Martines Sinues *et al* (2013)	Low	High	Low	Low		Low	Low	Low
Shafiek *et al* (2015)	Low	High	Low	Low		Low	Low	Low
Cazzolla *et al* (2015)	Low	Unclear	Low	Low		Low	Low	Low
Besa *et al* (2015)	Low	Unclear	Low	Low		Low	Low	Low
Gaida *et al* (2016)	Unclear	Unclear	Low	Low		Low	Low	Low
Allers *et al* (2016)	Low	Low	Low	Low		Low	Low	Low
Jerno-estaban *et al* (2017)	Low	High	Low	Low		Low	High	Low
Pizzini *et al* (2018)	Low	Low	Low	Low		Low	Low	Low
Bregy *et al* (2018)	High	High	Unclear	Low		Unclear	Low	Unclear
Scarlata *et al* (2018)	High	High	Low	Low		Low	Low	Low
Phillips *et al* (2014)	Low	Low	Low	Low		Low	Low	High
Incalzi *et al* (2012)	Low	High	Low	Low		Low	Low	Low
Rodriguez-Aguilar *et al* (2019)	Low	Unclear	Low	Low		Low	Low	Low
Velzen *et al* (2019)	Unclear	Low	Low	Low		Low	Low	Low
Gaugg *et al* (2019)	High	High	Low	Low		Low	Low	Low

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Figure A: Risk of bias and applicability concerns graph: review authors' judgements about each domain presented as percentages across included studies



112 **Table B: Description of modification of QUADAS-2:**

Risk of Bias	QUADAS-2	Modified QUADAS-2
Patient selection	Was a consecutive or random sample of patients enrolled?	Was patient selection representative of the intended population
	Was a case-control design avoided?	Did the study include disease state and healthy controls
	Did the study avoid inappropriate exclusions?	Did the study avoid inappropriate exclusions?
Index test	Were the index test results interpreted without knowledge of the results of the reference standard?	Was the index test and data interpretation completed in a standardised and reproducible way?
	If a threshold was used, was it pre-specified?	Was any biomarker validation performed (internal or external)?
Reference standard	Is the reference standard likely to correctly classify the target condition?	Is the reference standards likely to correctly classify the target condition?
	Were the reference standard results interpreted without knowledge of the results of the index test?	Removed as not applicable
Flow and timing	Was there an appropriate interval between index test and reference standard?	Was there an appropriate interval between index test and reference standard?
	Did all patients receive the same reference standard?	Did all patients receive the same reference standard?
	Were all patients included in the analysis?	Were all patients included in the analysis?

Applicability	**QUADAS-2**	**Modified QUADAS-2**
Patient selection	Are there concerns that the included patients and setting do not match the review question?	Are there concerns that the included patients and setting do not match the review question?
Index test	Are there concerns that the index test, its conduct, or interpretation differs from the review question?	Could the conduct or interpretation of the index test have introduced bias?
Reference standard	Are there concerns that the target condition as defined by the reference standard does not match the question?	Are there concerns that the target condition as defined by the reference standard does not match the question?

Index test: Exhaled breath analysis

Reference standard: Internationally accepted standard for diagnosing asthma and COPD (i.e. following GINA and GOLD guidelines)

Target condition: Asthma and COPD

Intended use of the index test: diagnostic

3. PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4-5
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	7-8
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7-8
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary material
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7-8

Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	7-8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7-8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	N/A

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8 and Supplementary material
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9 and Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8,9,10,11 and Tables 1-2
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	8 and supplementary material
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	9-11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	Supplementary material
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A

DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14-15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	18
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	19
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	Declarations

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: **www.prisma-statement.org**.

Compound and chemical classification	Author	Disease	Comments
1. Carbonyl containing			
1.1. Aldehydes			
2-oxoglutaric acid semi-aldehyde	Bregy *et al* [83]	COPD vs. controls	Analysis by SESI-MS - high levels in COPD patients
aspartic acid semi-aldehyde	Bregy *et al* [83]	COPD vs. controls	Analysis by SESI-MS - high levels in COPD patients
Benzaldehyde	Phillips *et al* [38]	COPD vs. controls	
Butanal	Rodriguez-Aguilar *et al [40]*	COPD vs. controls	Positively correlated to COPD
Decanal	Basanta *et al* [21]	COPD vs. controls	Identifying COPD (GC-TOF-MS)
Dodecanal	Basanta *et al* [21]	COPD vs. controls	Identifying COPD (GC-TOF-MS)
Hexanal	Basanta *et al* [21]	COPD vs. controls	Identifying COPD (GC-TOF-MS)
Hexanal	Jareno-estaban *et al* [46]	COPD vs. controls	Discriminates between COPD and healthy controls
Hexanal	Phillips *et al* [38]	COPD vs. controls	
Nonanal	Basanta *et al* [21]	COPD vs. controls	Identifying COPD (GC-TOF-MS)
Nonanal	Jareno-estaban *et al* [46]	COPD vs. controls	Discriminates smokers and former smokers (with and without COPD) and never smokers
Nonanal	Schliech *et al* [18]	Asthma	Discriminates paucigranulocytic and neutrophilic asthma (Higher in neutrophilic asthma)
Nonanal	Schliech *et al* [18]	Asthma	Discriminates eosinophilic and neutrophilic asthma (Higher in neutrophilic asthma)
Octanal	Brinkman *et al [75]*	Asthma	Association between exhaled breath VOCs and urinary levels of salbutamol and OCS

Pentadecanal	Basanta *et al* [21]	COPD vs. controls	Identifying COPD (GC-TOF-MS)
Pentadecanal	Ibrahim *et al* [15]	Asthma vs. controls	Negatively correlated to asthma
Undecanal	Basanta *et al* [21]	COPD vs. controls	Identifying COPD (GC-TOF-MS)
3-methyl-propanal	Rodriguez-Aguilar *et al [40]*	COPD vs. controls	Positively correlated to COPD
1.2. Esters			
Ethyl 2,2-dimethylacetoacetate	Ibrahim *et al* [15]	Asthma vs. controls	Positively correlated to asthma
Linalylacetate	Gaida *et al* [47]	COPD vs. controls	VOCs seem to be related to COPD
1.3. Ketones			
2-butanone	Allers *et al* [82]	COPD vs. controls	IMS - smoking related compounds
2-butanone	Ibrahim *et al* [15]	Asthma vs. controls	Positively correlated to asthma
2-hexanone	Schliech *et al* [18]	Asthma	Discriminates pauci-granulocytic and eosinophilic asthma (lower in eosinophilic asthma)
2-pentanone	Allers *et al* [82]	COPD vs. controls	Detected by GC-APCI-MS discriminates COPD from healthy volunteers
2-pentanone	Pizzini *et al* [32]	Acute and stable COPD vs. controls	Indicative of acute exacerbation of COPD (positive correlation)
4-heptanone	Pizzini *et al* [32]	Acute and stable COPD vs. controls	Indicative of acute exacerbation of COPD (positive correlation)
6-methyl-5-hepten-2-one	Pizzini *et al* [32]	Acute and stable COPD vs. controls	Non-specific. Significant difference between COPD and healthy volunteers (higher in healthy)
Acetone	Martines *et al* [80]	COPD vs. controls	Discriminate COPD from healthy volunteers using IMS
Cyclohexanone	Pizzini *et al* [32]	Acute and stable COPD vs. controls	Indicative of acute exacerbation of COPD (positive correlation)

Cyclohexanone (CAS 108-94-1)	Westhoff *et al* [76]	COPD vs. controls	IMS identified - raised in COPD patients
Cyclopentanone	Rodriguez-Aguilar *et al [40]*	COPD vs. controls	Positively correlated to COPD
1.4. Organic acids			
11-hydroxyundecanoic acid	Bregy *et al* [83]	COPD vs. controls	Analysis by SESI-MS - low levels in COPD patients
2-hydroxyisobutyric acid	Bregy *et al* [83]	COPD vs. controls	Analysis by SESI-MS - compound predictive that breath is from a COPD patient
2-methyl butanoic acid	Rodriguez-Aguilar *et al [40]*	COPD vs. controls	Negatively correlated to COPD
Acetic acid	Phillips *et al* [38]	COPD vs. controls	
Butanoic acid	Basanta *et al* [21]	COPD vs. controls	Identifying COPD (GC-TOF-MS) - negative correlation
Dodecanedioic acid	Bregy *et al* [83]	COPD vs. controls	Analysis by SESI-MS - low levels in COPD patients
Oxoheptadecanoic acid	Bregy *et al* [83]	COPD vs. controls	Analysis by SESI-MS - low levels in COPD patients
Pentanoic acid	Basanta *et al* [21]	COPD vs. controls	Identifying COPD (GC-TOF-MS) - negative correlation
Lysine	Brinkman *et al [75]*	Asthma	Association between exhaled breath VOCs and urinary levels of salbutamol and OCS
Glycolic acid	Brinkman *et al* [75]	Asthma	Association between exhaled breath VOCs and urinary levels of salbutamol and OCS
ω-oxo-alkenoic acids	Gaugg *et al* [33]	COPD	Levels significantly reduced in frequent COPD exacerbators
ω-hydroxy acids	Gaugg *et al [33]*	COPD	Levels significantly reduced in frequent COPD exacerbators
2. Hydrocarbons			
2.1. Alkanes			

3,7-dimethylnonane	Schliech *et al* [18]	Asthma	Discriminates eosinophilic and neutrophilic asthma (Higher in neutrophilic asthma) and lower eosinophilic asthma
Hexane	Schliech *et al* [18]	Asthma	Discriminates paucigranulocytic and eosinophilic asthma (lower in eosinophilic asthma)
Undecane	Schliech *et al* [18]	Asthma	Discriminates paucigranulocytic and neutrophilic asthma (Higher in paucigranulocytic asthma)
2,6-Dimethyl-heptane	Van Berkel *et al* [19]	COPD vs. controls	Classification model differentiated COPD from healthy volunteers
4,7-Dimethyl-undecane	Van Berkel *et al* [19]	COPD vs. controls	Classification model differentiated COPD from healthy volunteers
4-Methyl-octane	Van Berkel *et al* [19]	COPD vs. controls	Classification model differentiated COPD from healthy volunteers
Hexadecane	Van Berkel *et al* [19]	COPD vs. controls	Classification model differentiated COPD from healthy volunteers
6-ethyl-2-methyl-Decane	Cazzola *et al* [39]	COPD vs. controls	Positively correlated to COPD
Decane	Cazzola *et al* [39]	COPD vs. controls	Positively correlated to COPD
Hexane, 3-ethyl-4-methyl-	Cazzola *et al* [39]	COPD vs. controls	Negatively correlated to COPD
Tridecane	Gaida *et al* [47]	COPD vs. controls	VOCs seem to be related to COPD
Tetradecane	Rodriguez-Aguilar *et al [40]*	COPD vs. controls	Negatively correlated to COPD
2 methyl-decane	Ibrahim *et al* [15]	Asthma vs. controls	Positively correlated to asthma
2,6,10-trimethyl-dodecane	Ibrahim *et al* [15]	Asthma vs. controls	Positively correlated to asthma
2,6,11-trimethyl-dodecane	Ibrahim *et al* [15]	Asthma vs. controls	Positively correlated to asthma
5,5-Dibutylnonane	Ibrahim *et al* [15]	Asthma vs. controls	Negatively correlated to asthma
Pentane	Olopade *et al* [29]	Acute and stable asthma vs. controls	Raised in asthma

Ethane	Paredi *et al* [36]	Asthma vs. controls	Raised in asthma
Butane	Phillips *et al* [38]	COPD vs. controls	Undetermined correlation
2,4-dimethylheptane	Pizzini *et al* [32]	Acute and stable COPD vs. controls	Lower in COPD exacerbations compared to healthy volunteers
2,6-dimethyloctane	Pizzini *et al* [32]	Acute and stable COPD vs. controls	Lower in COPD exacerbations compared to healthy volunteers
2-methylhexane	Pizzini *et al* [32]	Acute and stable COPD vs. controls	Higher in COPD exacerbations compared to healthy volunteers
cyclohexane	Pizzini *et al* [32]	Acute and stable COPD vs. controls	Higher in COPD exacerbations compared to healthy volunteers
n-butane	Pizzini *et al* [32]	Acute and stable COPD vs. controls	Indicative of acute exacerbation of COPD (negative correlation)
n-Heptane	Pizzini *et al* [32]	Acute and stable COPD vs. controls	Indicative of stable COPD (positive correlation)
2.2. Alkenes			
3- tetradecene	Schliech *et al* [18]	Asthma	Discriminates paucigranulocytic and neutrophilic asthma (Higher in neutrophilic asthma)
Pentadecene	Schliech *et al* [18]	Asthma	Discriminates paucigranulocytic and neutrophilic asthma (Higher in neutrophilic asthma)
Isoprene	Van Berkel *et al* [19]	COPD vs. controls	Classification model differentiated COPD from healthy volunteers
1-Pentene, 2,4,4-trimethyl-	Cazzola *et al* [39]	COPD vs. controls	Negatively correlated to COPD
1,6-Dimethyl-1,3,5-heptatriene	Gaida *et al* [47]	COPD vs. controls	VOCs seem to be related to COPD
3,5-heptatriene	Gaida *et al* [47]	COPD vs. controls	VOCs seem to be related to COPD
Isoprene	Phillips *et al* [38]	COPD vs. controls	Differentiating stable COPD patients

2.3. Hydrocarbons			
C16 hydrocarbon	Van Berkel *et al* [19]	COPD vs. controls	Classification model differentiated COPD from healthy volunteers
4-ethyl-o-xylene	Ibrahim *et al* [15]	Asthma vs. controls	Negatively correlated to asthma
Isoprene	Lastard *et al* [67]	Asthma vs. controls	Low in asthmatics
Nonadecane	Phillips *et al* [38]	COPD vs. controls	Undetermined correlation
Octane	Rodriguez-Aguilar *et al [40]*	COPD vs. controls	Positively correlated to COPD
2.4. Monoaromatics			
Benzene, 1,3,5-tri-tert-butyl-	Cazzola *et al* [39]	COPD vs. controls	Negatively correlated to COPD
1-Ethyl-3-methyl benzene	Gaida *et al* [47]	COPD vs. controls	VOCs seem to be related to COPD
m/p-Xylene	Gaida *et al* [47]	COPD vs. controls	VOCs seem to be related to COPD
O-xylene	Gaida *et al* [47]	COPD vs. controls	VOCs seem to be related to COPD
Benzene	Phillips *et al* [38]	COPD vs. controls	Undetermined correlation
Toluene	Phillips *et al* [38]	COPD vs. controls	Undetermined correlation
2.5. Terpenes			
Limonene	Cazzola *et al* [39]	COPD vs. controls	Negatively correlated to COPD
Terpinolene	Ibrahim *et al* [15]	Asthma vs. controls	Positively correlated to asthma
4-Carene	Brinkman *et al [75]*	Asthma	Association between exhaled breath VOCs and urinary levels of salbutamol and OCS
3. Alcohol and Phenols			

3.1. Alcohols			
Cyclohexanol	Basanta *et al* [21]	COPD vs. controls	Identifying COPD (GC-TOF-MS)
Bicyclo[2.2.2]octan-1-ol, 4-methyl - C9H16O	Brinkman *et al* [25]	Acute and stable asthma vs. controls	Correlated with sputum eosinophils during loss of asthma control and with FENO during loss of asthma control
Methanol CH3OH	Brinkman *et al* [25]	Acute and stable asthma vs. controls	Correlated with FEV1% predicted during loss of asthma control
2-Propanol	Cazzola *et al* [39]	COPD vs. controls	Negatively correlated to COPD
2-Propanol	Rodriguez-Aguilar *et al [40]*	COPD vs. controls	Positively correlated to COPD
Phenole	Gaida *et al* [47]	COPD vs. controls	VOCs seem to be related to COPD
2-butylcyclohexanol	Ibrahim *et al* [15]	Asthma vs. controls	Negatively correlated to asthma
2-butyloctanol	Rodriguez-Aguilar *et al* [40]	COPD vs. controls	Positively correlated to COPD
Benzyl alcohol	Ibrahim *et al* [15]	Asthma vs. controls	Positively correlated to asthma
Phenol	Phillips *et al* [38]	COPD vs. controls	Undetermined correlation
1-propanol	Schliech *et al* [18]	Asthma	Discriminates eosinophilic and neutrophilic asthma (Higher in neutrophilic asthma) and lower in eosinophilic asthma
3.2. Phenol derivatives			
Butylated hydroxytoluene	Cazzola *et al* [39]	COPD vs. controls	Negatively correlated to COPD
m/p-Cresol,	Gaida *et al* [47]	COPD vs. controls	VOCs seem to be related to COPD
4. Others			

4.1.Sulphides			
Phthalic anhydride	Phillips *et al* [38]	COPD vs. controls	Undetermined correlation
Sulphur dioxide	Phillips *et al* [38]	COPD vs. controls	Undetermined correlation
dimethyl disulfide	Pizzini *et al* [32]	Acute and stable COPD vs. controls	Non-specific. Significant difference between COPD and healthy volunteers (higher in healthy)
methyl propyl sulfide	Pizzini *et al* [32]	Acute and stable COPD vs. controls	Indicative of stable COPD (positive correlation)
4.2. Permanent gases			
Ethyl 4-nitrobenzoate	Ibrahim *et al* [15]	Asthma vs. controls	Negatively correlated to asthma
Indole	Martines *et al* [80]	COPD vs. controls	Discriminate COPD from healthy volunteers using IMS
Indole	Gaida *et al* [47]	COPD vs. controls	VOCs seems to be related to COPD
4.3. Heterocycles			
Oxirane-dodecyl	Basanta *et al* [21]	COPD vs. controls	VOCs seem to be related to COPD
γ-hydroxy-L-homoarginine	Bregy *et al* [83]	COPD vs. controls	Analysis by SESI-MS - compound predictive that breath is from a COPD patient
4.4. Nitriles			
Ace-tonitrile - C2H3N	Brinkman *et al* [25]	Acute and stable asthma vs. controls	Correlated with sputum eosinophils and FEV1% during loss of asthma control
Hexyl ethylphosphonofluoridate	Cazzola *et al* [39]	COPD vs. controls	Negatively correlated to COPD
4.5. Anhydrides			
Acetonitrile	Allers *et al* [82]	COPD vs. controls	IMS - smoking related compound
4.6. Furans			

2-pentylfuran	Basanta *et al* [21]	COPD vs. controls	Identifying COPD (GC-TOF-MS)
4.7. Quinones			
Carbon dioxide	Phillips *et al* [38]	COPD vs. controls	Undetermined correlation
4.8.Others			
2,6-Di-tert-butylquinone	Ibrahim *et al* [15]	Asthma vs. controls	Negatively correlated to asthma
3,4-Dihydroxybenzonitrile	Ibrahim *et al* [15]	Asthma vs. controls	Positively correlated to asthma
Allyl methyl sulphide	Ibrahim *et al* [15]	Asthma vs. controls	Positively correlated to asthma

Table C: Table outlining reported VOC biomarkers in Asthma and COPD.